

Yes. For **Task 3**, a strong and sensible set for your brain MRI dataset is:

* **SimCLR**

* **BYOL**

* **MAE**

This trio gives you three different SSL ideas:

- * learn by **comparing two views of the same image**,
- * learn by **matching a student to a slowly updated teacher**,
- * learn by **rebuilding missing parts of an image**. ([arXiv][1])

Below is a **beginner-friendly narrative** you can directly use for your report or convert into **6 to 8 slides**.

Task 3: Select and Explain Three SSL Methods

Why SSL makes sense for this dataset

Self-supervised learning is useful when you have many images and want the model to learn useful patterns **before** using labels. Your dataset is made of brain MRI images, which means the model should learn texture, shape, edges, brightness changes, and abnormal regions. SSL is a good fit here because it can first learn the visual structure of MRI scans, then later use that knowledge for classification. ([arXiv][1])

1) SimCLR

Simple idea

SimCLR teaches the model like this:

> “These two changed versions of the **same MRI image** should still mean the same thing.”

For example, if we take one MRI image and make:

- * one slightly cropped version,
- * one slightly brighter or flipped version,

the model should understand that both still come from the same original image. At the same time, it should keep them different from other images in the batch. SimCLR is a contrastive method built around strong augmentations, a projection head, and a contrastive loss. ([arXiv][1])

Friendly diagram

```text

Original MRI

|  
|----> Augmented View 1 ----> Encoder ----> Vector A  
|

|----> Augmented View 2 ----> Encoder ----> Vector B

Goal:

Vector A and Vector B should be close

Vectors from other images should be far

...

### Positives and negatives

\* \*\*Positive pair\*\* = two augmented versions of the \*\*same\*\* image

\* \*\*Negative pairs\*\* = augmented views from \*\*different\*\* images in the batch

This is the main idea of contrastive learning in SimCLR. ([arXiv][1])

### Why SimCLR fits brain MRI data

It is a good choice because MRI classes often differ through visual structure and region patterns. SimCLR helps the encoder learn meaningful image features without labels. It is also one of the most standard and well-known SSL baselines, so it is easy to justify academically. One practical caution is that augmentation choice matters a lot, and medical images need safer augmentations than natural photos. ([arXiv][1])

### One-line classroom explanation

SimCLR is like showing the model the same brain scan in two slightly different forms and saying,

\*\*“These are still the same scan. Learn what stays important.”\*\*

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## ## 2) BYOL

### ### Simple idea

BYOL is similar to SimCLR in using two augmented views of the same image, but it does **not** rely on negative pairs. Instead, it uses:

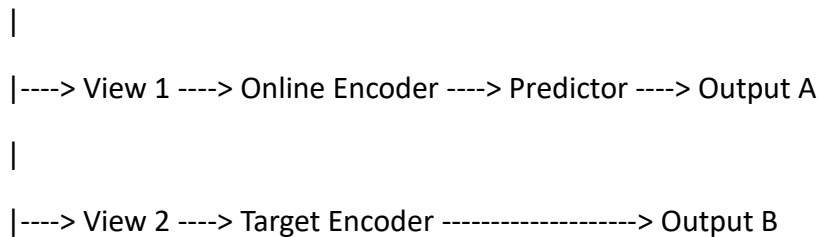
- \* an **online network** that learns,
- \* a **target network** that changes slowly.

The online network tries to predict the representation made by the target network for another view of the same image. BYOL's key idea is that useful representations can be learned without explicit negative samples. ([arXiv][2])

### ### Friendly diagram

```text

Original MRI



Goal:

Output A should match Output B

Target network updates slowly over time

...

Positives and negatives

* **Positive relationship** = two views of the same image

* **Negative pairs** = not explicitly used

That is what makes BYOL different from SimCLR. ([arXiv][2])

Why BYOL fits brain MRI data

BYOL can be attractive for MRI because medical datasets often do not behave like very large natural-image datasets, and choosing strong negative pairs is not always ideal. Since BYOL avoids explicit negatives, it can be a cleaner option when images are visually similar and fine-grained. It is also a good conceptual contrast to SimCLR because both use two views, but BYOL removes the negative-pair part. ([arXiv][2])

One-line classroom explanation

BYOL is like a student learning from a calm teacher version of itself:

“Try to make your answer match the teacher’s answer for the same scan.”

3) MAE

Simple idea

MAE stands for **Masked Autoencoder**. Instead of comparing two views, it hides many parts of the image and asks the model to reconstruct what is missing. In the original MAE design, a large portion of image patches, often around **75%**, is masked, and the model learns from the visible patches to rebuild the hidden ones. ([arXiv][3])

Friendly diagram

```text

Original MRI

|

|----> Split into patches

|----> Hide many patches

|

Visible patches ----> Encoder ----> Decoder ----> Rebuild missing patches

Goal:

Predict the hidden parts correctly

...

### ### Masking / tokenization

This method uses **patches**:

- \* the image is cut into small square parts,

- \* many patches are hidden,

- \* the model sees only some patches,
- \* then it tries to reconstruct the missing ones.

This is different from SimCLR and BYOL because MAE learns from a **reconstruction task**, not from pair matching. ([arXiv][3])

### ### Why MAE fits brain MRI data

MRI images contain strong spatial structure. If the model learns to fill in missing parts, it may become better at understanding:

- \* local texture,
- \* tissue boundaries,
- \* symmetry,
- \* suspicious regions.

MAE is especially natural when using Vision Transformer-style backbones, because it works directly with image patches. ([arXiv][3])

### ### One-line classroom explanation

MAE is like giving the model an incomplete brain scan and asking:

**“Can you guess the missing parts by understanding the whole picture?”**

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# Why these three are a good set together

These three methods are a strong group because each one teaches the model in a different way:

- \* **SimCLR** learns by **pulling same-image views together and pushing other images apart**

- \* **BYOL** learns by **matching a student network to a slowly updated teacher**

- \* **MAE** learns by **reconstructing hidden image patches** ([arXiv][1])

So together they give you:

- \* one **contrastive** method,

- \* one **negative-free bootstrap** method,

- \* one **reconstruction-based** method. ([arXiv][1])

That makes your Task 3 answer stronger, because you are not choosing three methods that all work in the exact same way.

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# Best short justification for your dataset

For a brain MRI classification project, these three methods are suitable because MRI images depend heavily on structure, texture, and region consistency rather than colorful object backgrounds. SimCLR can learn stable visual similarity under safe augmentations, BYOL can learn strong features without explicit negatives, and MAE can learn spatial understanding by reconstructing masked regions. That makes them well matched to grayscale-like medical image patterns and useful as SSL candidates before downstream tumor classification. ([arXiv][1])

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## # Ready-made slide outline

### ## Slide 1: What is SSL?

- \* SSL means learning from images **\*\*without using labels first\*\***
- \* The model creates its own learning task
- \* Useful for image datasets like brain MRI

### ## Slide 2: Why SSL for brain MRI?

- \* MRI images have important texture and shape patterns
- \* Labels may be limited or expensive
- \* SSL helps learn strong features before classification

### ## Slide 3: SimCLR

- \* Two views of same image
- \* Positive pair = same image
- \* Negative pairs = other images
- \* Good for learning similarity

### ## Slide 4: BYOL

- \* Two views of same image
- \* Online network learns from target network
- \* No explicit negatives
- \* Good when negative-pair setup is not ideal

### ## Slide 5: MAE

- \* Hide many image patches
- \* Model rebuilds missing patches
- \* Helps learn image structure and context

### ## Slide 6: Why these three?

- \* SimCLR = contrastive
- \* BYOL = teacher-student, no negatives
- \* MAE = reconstruction
- \* Together they cover three important SSL families

### ## Slide 7: Which may fit best?

- \* SimCLR: strong baseline
- \* BYOL: simple and powerful without negatives
- \* MAE: strong for patch-based image understanding
- \* Final best method should be decided experimentally

### ## Slide 8: References

\* SimCLR paper

\* BYOL paper

\* MAE paper

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## # References

\* Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. "A Simple Framework for Contrastive Learning of Visual Representations." 2020. ([arXiv][1])

\* Grill, J.-B. et al. "Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning." 2020. ([arXiv][2])

\* He, K. et al. "Masked Autoencoders Are Scalable Vision Learners." 2021. ([arXiv][3])

If you want, I can turn this into a **ready-to-submit 6 to 8 slide presentation text** next.

[1]: [https://arxiv.org/abs/2002.05709?utm\\_source=chatgpt.com](https://arxiv.org/abs/2002.05709?utm_source=chatgpt.com) "A Simple Framework for Contrastive Learning of Visual Representations"

[2]: [https://arxiv.org/abs/2006.07733?utm\\_source=chatgpt.com](https://arxiv.org/abs/2006.07733?utm_source=chatgpt.com) "Bootstrap your own latent: A new approach to self-supervised Learning"

[3]: [https://arxiv.org/abs/2111.06377?utm\\_source=chatgpt.com](https://arxiv.org/abs/2111.06377?utm_source=chatgpt.com) "Masked Autoencoders Are Scalable Vision Learners"